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Influence of polymer amphiphilicity on the cyclization efficiency of poly (2-oxazoline)s

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ABSTRACT

The steadily increasing interest in cyclic polymers has led to a plethora of proposed applications, particularly as biomaterials. However, the large-scale synthesis of cyclic polymers remains one of the greatest challenges in the field. One reason for this is the poorly understood role of the polymer structure on the cyclization efficiency, leading to synthesis protocols that must be optimized on a case-by-case basis without consideration for the chemical nature of the polymer. Recently, we found through a combination of laboratory experiments and molecular dynamics simulations that the polymer backbone flexibility has a major influence on the cyclization efficiency of poly(cyclic imino ether)s. Concurrently, we identified that a more flexible polymer backbone was also better solvated in the given organic solvent. To deconvolute the importance of the polymer backbone flexibility and solvation, in this study a variety of α-alkyne, ω-azide terminated (co)polymers of 2-ethyl-2-oxazoline and 2–*n*-butyl-2-oxazoline with different amphiphilicities, but the same polymer backbone, have been synthesized and subjected to copper-catalysed azide-alkyne cycloaddition (CuAAC) reactions in water. In agreement with our previous studies, it was observed that an increase in polymer chain hydrophobicity, and hence a decrease in solvation, leads to a significant decrease in cyclization efficiency. To further understand the effect of different CuAAC reaction conditions on the reaction outcome, different polymer concentrations and reaction temperatures were screened. This study highlights the importance of polymer amphiphilicity, and more generally solvation, in CuAAC protocols aimed towards the synthesis of cyclic polymers.

1. Introduction

The unique material properties of cyclic polymers, when compared to linear polymers, include decreased hydrodynamic dimensions [1], increased blood circulation time [2] and thermal stability of micelles [3,4]. These properties have led to an ever-increasing interest in these materials as biomaterials, resulting in proposed applications as protein-repellent surfaces [5–8], gene delivery materials [9] and drug delivery systems [10,11]. However, the large-scale synthesis of pure cyclic polymers remains one of the greatest challenges in the field [12]. This is especially true for the unimolecular ring closure method, in which the linkable chemical moieties can be directly introduced onto the polymer chain ends either during synthesis or through post-polymerization modification. This simple introduction of the reactive groups makes the named approach attractive for the synthesis of cyclic polymers as it can be expanded to a variety of different polymer backbones, such as poly((meth)acrylate)s [11,13,14], poly(*N*-isopropylacrylamide) [15,16], poly(styrene) [17–19] or poly(cyclic imino ether)s [6,9,20–22].

However, a major limitation of the ring closure method is that the molecular weight of the pre-cyclized polymers usually needs to be below 20 kDa [23] in order to minimize interchain reactions during the cyclization process. Interchain reactions lead to the formation of chain-chain coupling products rather than the monocyclic species. These higher molecular weight side-products are difficult to remove from the final reaction mixture [24,25] and so attempts to avoid intermolecular reactions commonly rely on highly diluted reaction conditions [6,26] or high catalyst loadings [18,27]. This drawback can be somewhat mitigated through the use of techniques such as continuous photo-flow [28,29] to counteract the yield limitations associated with reactions under high dilution. However, interchain coupling can rarely be fully supressed, resulting in labour- and time-intensive manual work-up procedures to separate the pure cyclic compounds through techniques such as preparative size exclusion chromatography (SEC) [22,30].

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Despite the need for reaction optimization to obtain high cyclization yields, so far only a handful of studies have systematically investigated the influence of reaction conditions on the cyclization efficiency. In fact, in most cases optimization is limited to experimental parameters, such as polymer concentration [18], catalyst loading [20], addition methods of the reagents [21], reaction temperatures [27] or time [31], not considering the polymer structure itself. This results in synthetic protocols that cannot be directly translated between different polymer backbones and sometimes even within the same polymer class.

During the last two decades, poly(2-oxazoline)s (POx) and the structurally related poly(2-oxazine)s (POzi) have gained attention in the field of biomaterials due to their chemical versatility [32] and tuneable thermoresponsive properties [33,34] based on the wide range of possible substituents on the polymer chain. Applications include drug delivery [35], surface modification [36–38], tissue engineering [39,40] and biofabrication [41–43]. In the context of macrocyclization, the wide range of available substituents make POx and POzi attractive classes of polymers to investigate the influence of polymer structure on the cyclization reaction efficiency.

Indeed, using POx and POzi, we recently found that the polymer structure of these materials plays a significant part in the cyclization process [21,44]. Using heterotelechelic POx and POzi featuring azide and alkyne end-groups, which can be used to facilitate the copper-catalysed azide-alkyne cycloaddition (CuAAC) reaction in dichloromethane, cyclization efficiencies were determined for both materials under several different reaction conditions. It was found that POzi cyclized significantly more efficiently [21] compared to POx, independent of polymer concentration or molecular weight of the materials. Molecular dynamics [44] highlighted that the additional CH_2 group in the polymer backbone of POzi compared to POx results in a significantly increased flexibility of the polymer backbone in solution, allowing the chain ends of POzi to come into closer proximity of one another more frequently than for POx. We assume that this close proximity of the chain ends in solution is directly related to the cyclization efficiency. Furthermore, in the same computational study, we were able to show that the POzi backbone is significantly better solvated than that of POx, indicating that this increased solvation also contributes to the increased cyclization efficiency of these materials.

A recent study by Zhang *et al.* [45] further suggests that polymer solvation could be a powerful handle to control the cyclization efficiency of polymers. Using an anthracene-capped amphiphilic statistical copolymer made from oligo(ethylene glycol)- and dodecyl-substituted acrylate monomers, they were able to demonstrate full conversion of the linear precursor into the monocyclic species at polymer concentrations as high as 40 mg mL^{-1}. The amphiphilic nature of the polymer caused self-folding in water, encapsulating the anthracene end-groups into the hydrophobic pocket of the polymer, effectively hindering polymer end-groups of multiple polymer chains from reacting to form interchain-coupled side-products. Pure cyclic products were obtained via light-induced dimerization of the anthracene groups. Impressively, this approach was used with high molecular weight polymers (>200 kg mol^{-1}), which is beyond the chain length that can usually be cyclized using unimolecular ring closure techniques.

Interestingly, based on our studies on POx and POzi and the above experimental study, it appears that the solvation of polymers influences the cyclization efficiency differently depending on the system. In the water-based anthracene system of Zhang *et al.* [45], a reduction in polymer solvation and isolation of the hydrophobic end-groups resulted in full cyclization of the precursor polymer into the monocyclic species. A similar effect of the polymer solvation on the cyclization efficiency has been observed by the groups of Becer [27] and Matyjaszewski [46] for CuAAC reactions of hydrophobic homopolymers in organic solvents. Meanwhile, in our theoretical and experimental studies in dichloromethane, it was the system with the higher solvation that demonstrated better cyclization efficiency. However, it should be noted that in the latter case it is difficult to isolate the importance of polymer backbone flexibility and solvation on the reaction outcome.

Therefore, in the current study, we set out to isolate the effect of polymer amphiphilicity and hence solvation on the cyclization efficiency of POx. A variety of statistical (co)polymers of hydrophilic 2-ethyl-2-oxazoline (EtOx) and hydrophobic 2-*n*-butyl-2-oxazoline (BuOx) with varying monomer feed ratios have been synthesized using cationic ring opening polymerization (CROP) and evaluated for their cyclization efficiency under different CuAAC reaction conditions. All polymers were modified with an alkyne group on the α- and an azide moiety on the ω-terminus of the polymer to facilitate intramolecular cyclization or intermolecular oligomerization through CuAAC reaction conditions in deionized water. In contrast to the previously mentioned CuAAC studies of fully hydrophobic polymers in organic solvents, we chose water as a reaction medium to accentuate the hydrophobic interactions between the *n*–butyl substituents and hence the solvation differences between the different amphiphilic polymers. Furthermore, using this approach it was possible to compare our Cu(I)-reliant system with the light-initiated system by Zhang *et al.* [45], investigating the differences in cyclization chemistries and monomer feed ratios on the cyclization efficiency.

2. Materials and methods

All chemicals were used as received unless noted otherwise. 2-*n*-butyl-2-oxazoline was synthesized from ethanolamine and valeronitrile using a previously published procedure by Witte and Seeliger [47]. Sodium ascorbate, 2-ethyl-2-oxazoline (≥99 %), ethanolamine (≥99 %), valeronitrile (99.5 %), lithium bromide (99 %), dimethylacetamide (DMAc, >99.9 %), sodium azide (99 %), propargyl benzenesulfonate (96 %), calcium hydride and deuterated chloroform (≥99 %) were purchased from Sigma Aldrich. Diethyl ether (anhydrous, >99.5 %) was purchased from Ajax Finechem. Dichloromethane (≥99.8 %) was purchased from Thermo Fisher Scientific. Anhydrous copper(II) sulfate was purchased from ChemSupply. Dry acetonitrile was obtained from a solvent purification system (Innovation Technology). Propargyl benzenesulfonate was distilled under reduced pressure over calcium hydride and stored under vacuum at −20 °C until used. 2-ethyl-2-oxazoline and 2-*n*-butyl-2-oxazoline were distilled under reduced pressure over calcium hydride and stored under vacuum at room temperature until used. SnakeSkin™ dialysis tubing with a molecular weight cut-off (MWCO) of 3.5 kDa was purchased from Thermo Fisher Scientific.

2.1. Size exclusion chromatography (SEC)

The SEC measurements were conducted on a PSS SECurity^2 system consisting of a PSS SECurity Degasser, PSS SECurity TCC6000 Column Oven (60 °C), PSS GRAM Column Set (8x150 mm 10 μm Precolumn, 8x300 mm 10 μm Analytical Columns, 1000 Å, 1000 Å and 30 Å) and an Agilent 1260 Infinity Isocratic Pump, Agilent 1260 Infinity Standard Autosampler, Agilent 1260 Infinity Diode Array and Multiple Wavelength Detector (A: 254 nm, B: 360 nm), Agilent 1260 Infinity Refractive Index (RI) Detector (35 °C). HPLC grade DMAc + 0.08 % LiBr, is used as eluent at a flow rate of 1 mL·min^{-1}. Narrow disperse linear poly(styrene) (M_n: 266 g mol^{-1} to 2.52×10^6 g mol^{-1}) and poly(methyl methacrylate) (M_n: 202 g mol^{-1} to 2.2×10^6 g mol^{-1}) standards (PSS ReadyCal) were used as calibrants. All samples were passed over 0.22 μm PTFE membrane filters. Molecular weight and dispersity analysis was performed in PSS WinGPC UniChrom software (version 8.33).

2.2. Nuclear magnetic resonance (NMR) spectroscopy

^{1}H NMR spectra were recorded on a Bruker System 600 Ascend LH, equipped with a BBO-Probe (5 mm) with z-gradient (^{1}H: 600.13 MHz). Resonances are reported in parts per million (ppm) relative to tetramethyl silane (TMS). The δ–scale was calibrated to the respective solvent signal of $CHCl_3$ in ^{1}H spectra.

2.3. Infrared (IR) spectroscopy

Infrared spectra were recorded on a Bruker Alpha-P Fourier Transform Infrared (FTIR) spectrometer equipped with a diamond crystal attenuated total reflectance (ATR) accessory. Spectra were obtained between 4000 and 400 cm^{-1}, as an average of 32 scans per sample.

2.4. Dynamic light scattering (DLS)

Dynamic Light Scattering experiments were conducted on a Malvern Nano-S Nano series Zetasizer. For temperature sweeps, the sample temperature was gradually increased in 2 °C increments and the temperature equilibrated at each temperature for 120 s. Data was collected at a scattering angle of 173° using polymer concentrations of either 5, 10 or 20 mg mL^{-1} in deionized water. Each sample was measured in triplicates.

3. Synthetic procedures

3.1. Polymerizations

A monomer mixture containing a feed ratio [M1]:[M2] (with M1 = EtOx and M2 = BuOx) of 50:0, 45:5, 40:10 or 35:15 eq. was prepared in acetonitrile at a total monomer concentration of 4 M. Propargyl benzenesulfonate (1 eq.) was added to the monomer mixture and the reaction stirred overnight at 70 °C. After full monomer conversion was confirmed using ^{1}H NMR experiments, the reaction was cooled to room temperature before dilution to 2 M using acetonitrile and the addition of sodium azide (10 eq.). After stirring for 24 h at room temperature, residual sodium azide was removed via centrifugation, the solvent evaporated, and the crude polymer was redissolved in a small amount of dichloromethane. After precipitation in cold anhydrous diethyl ether, the precipitate was redissolved in deionized water and lyophilized. The final product was obtained as a white powder.

PEtOx$_{50}$ (**P1**): ^{1}H NMR (600 MHz, $CDCl_3$) δ = 4.35 – 4.05 (br, 2H, HC≡CC$\underline{H}_2$), 3.44 (br, 238H, NC$\underline{H}_2$C$\underline{H}_2$), 2.65–2.90 (br, 1H, $\underline{H}$C≡CCH$_2$), 2.33 (br, 118H, (C=O)C$\underline{H}_2$CH$_3$), 1.11 (br, 178H, (C=O)CH$_2$C$\underline{H}_3$) ppm; SEC (DMAc, PMMA cal.): M_n = 7,200 g mol^{-1}, $Đ$ = 1.08; FTIR: ν = 2103 cm^{-1} (N_3). Yield: 1.83 g (92 %).

PEtOx$_{45}$-*stat*-BuOx$_5$ (**P2**): ^{1}H NMR (600 MHz, $CDCl_3$) δ = 4.35 – 4.05 (br, 2H, HC≡CC$\underline{H}_2$), 3.45 (br, 257H, NC$\underline{H}_2$C$\underline{H}_2$), 2.34 (br, 130H, (C=O)C$\underline{H}_2$CH$_2$CH$_2$CH$_3$, (C=O)C$\underline{H}_2$CH$_3$), 1.58 (br, 14H, (C=O)CH$_2$C$\underline{H}_2$CH$_2$CH$_3$), 1.33 (br, 14H, (C=O)CH$_2$CH$_2$C$\underline{H}_2$CH$_3$), 1.11 (br, 172H, (C=O)CH$_2$C$\underline{H}_3$), 0.91 (br, 21H, (C=O)CH$_2$CH$_2$CH$_2$C$\underline{H}_3$) ppm; SEC (DMAc, PMMA cal.): M_n = 7,800 g mol^{-1}, $Đ$ = 1.08; FTIR: ν = 2098 cm^{-1} (N_3). Yield: 1.85 g (92 %).

PEtOx$_{40}$-*stat*-BuOx$_{10}$ (**P3**): ^{1}H NMR (600 MHz, $CDCl_3$) δ = 4.35 – 4.05 (br, 2H, HC≡CC$\underline{H}_2$), 3.45 (br, 245H, NC$\underline{H}_2$C$\underline{H}_2$), 2.34 (br, 124H, (C=O)C$\underline{H}_2$CH$_2$CH$_2$CH$_3$, (C=O)C$\underline{H}_2$CH$_3$), 1.58 (br, 24H, (C=O)CH$_2$C$\underline{H}_2$CH$_2$CH$_3$), 1.33 (br, 25H, (C=O)CH$_2$CH$_2$C$\underline{H}_2$CH$_3$), 1.11 (br, 147H, (C=O)CH$_2$C$\underline{H}_3$), 0.91 (br, 38H, (C=O)CH$_2$CH$_2$CH$_2$C$\underline{H}_3$) ppm; SEC (DMAc, PMMA cal.): M_n = 7,600 g mol^{-1}, $Đ$ = 1.08; FTIR: ν = 2098 cm^{-1} (N_3). Yield: 1.82 g (91 %).

PEtOx$_{35}$-*stat*-BuOx$_{15}$ (**P4**): ^{1}H NMR (600 MHz, $CDCl_3$) δ = 4.35 – 4.05 (br, 2H, HC≡CC$\underline{H}_2$), 3.44 (br, 247H, NC$\underline{H}_2$C$\underline{H}_2$), 2.30 (br, 128H, (C=O)C$\underline{H}_2$CH$_2$CH$_2$CH$_3$, (C=O)C$\underline{H}_2$CH$_3$), 1.57 (br, 39H, (C=O)CH$_2$C$\underline{H}_2$CH$_2$CH$_3$), 1.33 (br, 39H, (C=O)CH$_2$CH$_2$C$\underline{H}_2$CH$_3$), 1.11 (br, 129H, (C=O)CH$_2$C$\underline{H}_3$), 0.91 (br, 57H, (C=O)CH$_2$CH$_2$CH$_2$C$\underline{H}_3$) ppm; SEC (DMAc, PMMA cal.): M_n = 7,100 g mol^{-1}, $Đ$ = 1.07; FTIR: ν = 2103 cm^{-1} (N_3). Yield: 1.74 g (87 %).

3.2. CuAAC reaction protocol

In a 25 mL round-bottom flask, the heterotelechelic precursor polymer was dissolved in deionized water (10 mL) at a concentration of either 5, 10 or 20 mg mL^{-1}. Sodium ascorbate (0.04 g, 0.2 mmol, 20 mM in solution) was added, and the reaction mixture was degassed with argon for 30 min. For experiments at elevated temperatures, the reaction mixture was then heated to either 32, 38 or 50 °C and stirred for an additional 30 min to achieve an even temperature profile throughout the solution. To initiate CuAAC, copper(II) sulfate (0.02 g, 0.1 mmol, 10 mM in solution) was added, and the reaction mixture was stirred for 24 h. After that, solids within the reaction mixture were removed via centrifugation before transferring the supernatant into a dialysis tube (MWCO: 3.5 kDa). In elevated temperature experiments where the polymer precipitated from solution, the polymer was resolubilized by cooling the reaction mixture in an ice bath prior to centrifugation. The sample was dialysed for 24 h against deionized water and lyophilized to yield an off-white powder.

4. Results and discussion

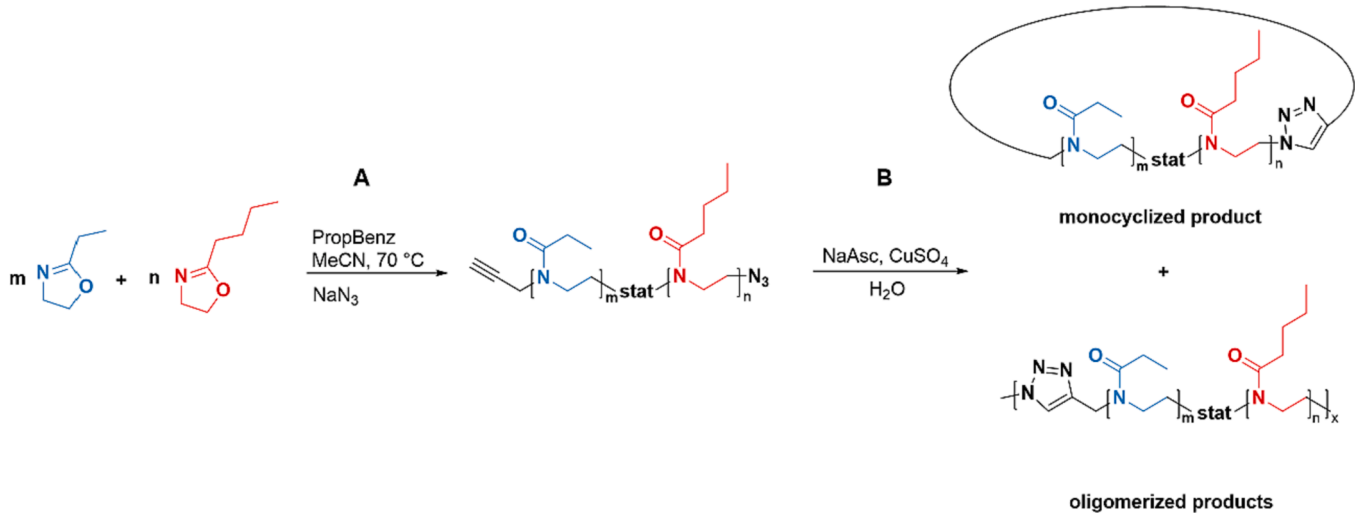
4.1. Polymer synthesis and characterization

Well-defined amphiphilic statistical (co)polymers of hydrophilic EtOx and hydrophobic BuOx monomers were synthesized using CROP at 70 °C in acetonitrile (MeCN, Scheme 1, A). The relatively low reaction temperature was chosen to minimize side-reactions during polymerization [48]. By adjusting the feed ratio of the monomers, (co)polymers containing either 0, 10, 20 or 30 mol% of BuOx and a total degree of polymerization of 50 were obtained. All polymerizations were initiated using propargyl benzenesulfonate (PropBenz) to introduce an alkyne group on the α-terminus of the polymers and terminated using sodium azide (NaN_3) to introduce an azide moiety on the ω-terminus once full monomer consumption was observed in the ^{1}H NMR spectra.

The obtained linear polymers **P1-P4** showed consistently low dispersities and monomodal characteristics in the SEC elugrams (Fig. S1), implying well-defined polymers and confirming the living character of the polymerizations (Table 1). The incorporation of BuOx in all cases was close to the predicted theoretical value, as determined through comparison of the proton signal integrals of the ethyl and *n*-butyl substituents in ^{1}H NMR spectra. Successful inclusion of the alkyne α-terminus was confirmed via the peaks in the ^{1}H NMR spectra between 4.05 and 4.35 ppm (Fig. S2-S5). Additionally, the sharp signal at around 2100 cm^{-1} in the IR spectra (Fig. S6) provides qualitative evidence of successful incorporation of the azide group at the ω-terminus of the polymers.

Independent of the BuOx content, the molecular weights of all polymers, as determined using ^{1}H NMR end group analysis, were higher than the theoretical values. This discrepancy can likely be attributed to minor ambient-light induced degradation of the PropBenz initiator, leading to an increase of the monomer-to-initiator ratio during polymerization and ultimately an increased molecular weight of the resulting polymer chains. This increase in molecular weight was even more pronounced in the SEC elugrams, which is caused by the use of a PMMA calibration for SEC analysis. As all SEC traces are monomodal and the dispersities remain below 1.10, it can be assumed that only little side reactions are present during polymerization, indicating that the end-group fidelity on the α-terminus has not been affected by the molecular weight increase.

To demonstrate the different amphiphilicities of the polymers, the thermoresponsive properties were evaluated using DLS over a range of temperatures (Fig. 1). The thermoresponsive properties of a given polymer at a distinct concentration in solution can be characterized by its cloud point temperature (T_{CP}). The T_{CP} describes the temperature at which the favourable enthalpy of mixing term, caused by hydrogen bonds between water molecules and the polymer chains at low temperatures, is overcome by the increasing entropic T·ΔS term of the free energy of mixing equation. This results in an overall positive value for the free energy of mixing, dehydration of the polymer chains and ultimately polymer aggregation and precipitation out of solution [49]. As a result, the polymer solution turns from clear to turbid, and the increased



Scheme 1. (A) The cationic ring opening polymerization of EtOx (blue) and BuOx (red) in acetonitrile (MeCN). The polymerization was initiated using propargyl benzenesulfonate (PropBenz) with sodium azide (NaN_3) serving as the terminating agent. (B) The obtained heterotelechelic polymers were then used in copper-catalysed azide-alkyne cycloaddition (CuAAC) reactions using sodium ascorbate (NaAsc) and copper(II) sulfate ($CuSO_4$) as a catalyst system to form product mixtures of either the monocyclized polymer or oligomeric products, formed through interchain coupling reactions.

Table 1
BuOx content of (co)polymers from EtOx and BuOx, number averaged molecular weight (M_n), dispersities (Đ) and cloud point temperatures (T_{CP}) of the synthesized (co)polymers **P1**-**P4** before cyclization.

Polymer	BuOx content (theo.) [mol%]	BuOx content[a] [mol%]	M_n (theo.) [kg mol^{-1}]	M_n^b [kg mol^{-1}]	M_n^c [kg mol^{-1}]	Đ^c [-]	T_{CP}^d [°C]
P1	0	0	5.0	5.8	7.2	1.08	n/a
P2	10	11	5.1	6.5	7.8	1.08	64
P3	20	21	5.2	6.4	7.6	1.08	44
P4	30	31	5.4	6.7	7.1	1.07	30

[a] Determined through comparison of the proton signal integrals of the ethyl and *n*-butyl substituents in ^{1}H NMR spectra. [b] Determined using end-group analysis via ^{1}H NMR spectroscopy. [c] Determined using SEC with dimethylacetamide (DMAc) + 0.08 % LiBr as an eluent against PMMA standards. [d] Determined using DLS at a polymer concentration of 5 mg mL^{-1} in deionized water.

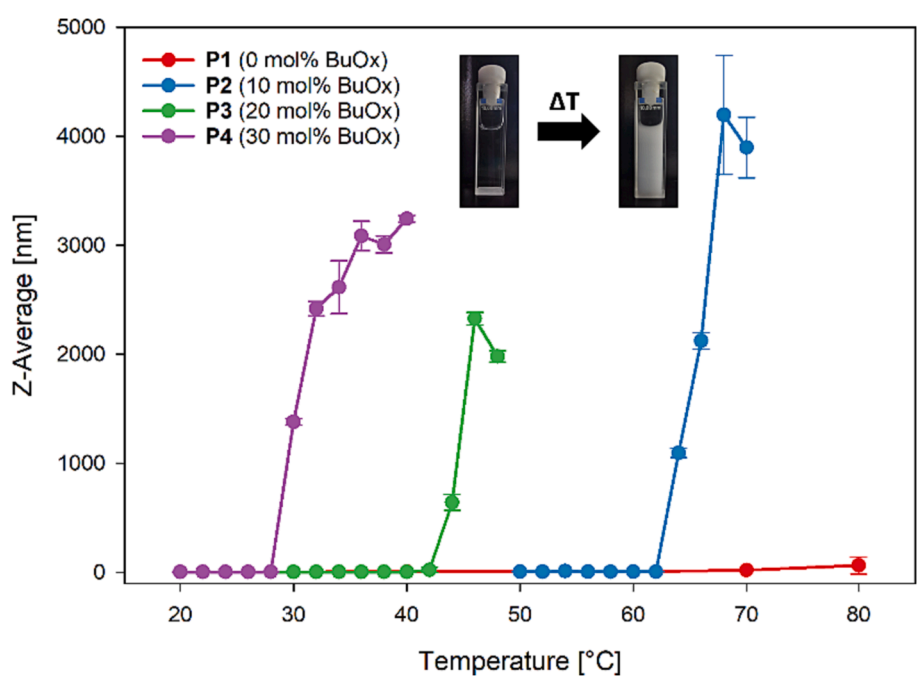


Fig. 1. Z-average of **P1**-**P4** before cyclization as a function of temperature to determine the T_{CP} at a polymer concentration of 5 mg mL^{-1} in deionized water. Data was obtained using DLS experiments at a scattering angle of 173°. Each polymer sample was measures as a triplicate.

particle size of the polymer aggregates can be detected using DLS. For the current study, the T_{CP} was determined as the first temperature at which particles with a size of ≥ 30 nm formed. As the presence of hydrophobic BuOx units within the polymer chain favours hydrophobic interactions, it is expected that an increasing content of BuOx monomers in the polymer chain will reduce the solvation in water and lead to decreased T_{CP} values.

Unsurprisingly, **P1**, which is composed of only EtOx units, showed no T_{CP} and only a marginal increase in overall particle size up to a temperature of 80 °C, indicating good solvation of this polymer in water. As the BuOx content was increased to 10 mol% (**P2**), the overall hydrophilicity of the polymer chain decreases, leading to an observable T_{CP} at 64 °C. This value decreased further to 44 °C and finally 30 °C, as the BuOx content increased to 20 (**P3**) and 30 mol% (**P4**), respectively. Synthesizing copolymers with BuOx content higher than 30 mol% was not attempted as the T_{CP} would be below room temperature and thus any CuAAC cyclization reaction would require cooling in order to keep the polymer in solution.

It should be noted that the thermoresponsive polymers **P2-P4** all showed a sharp increase of particle size over a small temperature range, which is common for well-defined POx in solution [50] and further highlights the successful synthesis of these materials. Additionally, the maximum particle size above T_{CP} seems to be dependent on the BuOx content in the polymers. This might be a result of the different ratios between hydrophilic and -phobic units within the polymer chains. Out of all thermoresponsive polymers, **P2** showed the largest particle size of >4.2 μm, potentially because of the low amount of BuOx in the polymer chain, which prevents the formation of fully collapsed globules in solution, and hence leads to less compact aggregates above the T_{CP}. As the BuOx content was increased to 20 mol% (**P3**), the particle size above the T_{CP} decreased significantly to a maximum 2.3 μm, indicating better compaction of the aggregates. Surprisingly, the particle size increased to 3.2 μm for **P4**, which may be rooted in higher aggregation tendency of the polymer chains in solution at 30 mol% BuOx, leading to overall larger polymer aggregates. The significant changes in T_{CP} between the polymers before cyclization confirmed the different solvation of these systems in water, making them suitable to study the influence of polymer amphiphilicity and hence solvation on the cyclization efficiency.

4.2. *The influence of polymer amphiphilicity on cyclization efficiency*

To investigate the influence of polymer amphiphilicity on the cyclization efficiency of **P1-P4** the polymers were exposed to CuAAC reaction conditions to afford the cycloaddition between the alkyne and azide terminus of the polymer chains (Scheme 1, B). These reactions are often conducted in either pure organic solvents, such as dichloromethane [21,51] and *N,N*–dimethylformamide [52,53] or in mixtures of organic solvents and water, such as tetrahydrofuran/water [22]. This allows for adjustment of parameters such as reactant/catalyst solubility [54], reaction workup [21,55], or to exploit high solvent boiling points [56] in an effective manner. However, to accentuate the different solution properties of the polymers in this study, organic solvents were avoided, and water was chosen as the solvent. The catalysing Cu(I) species was generated *in situ* through reduction of copper(II) sulfate with sodium ascorbate (NaAsc). To counteract the disproportionation reaction of the formed Cu(I) species into Cu(0) and Cu(II) and ensure regeneration of Cu(I) ions, the concentrations of these compounds in the reaction mixture were fixed to 20 mM (NaAsc) and 10 mM ($CuSO_4$) (Fig. 2).

In all cases, the SEC elugrams show the formation of several different species upon CuAAC (Fig. 2, red). On the one hand, the signal of the linear heterotelechelic precursor polymers (Fig. 2, black) shifts towards higher elution volumes, indicating a reduction in hydrodynamic volume, which is associated with the formation of the monocyclic polymer topology [1]. Conversely, undesirable interchain coupling reaction products between polymer chains, macrocyclic or linear, can be identified through their higher hydrodynamic volumes in solution and hence a shift towards lower elution volumes in the SEC elugrams compared to the linear precursors. Since a direct comparison of the yield for the monocyclic product is difficult using the SEC elugrams alone, the yield

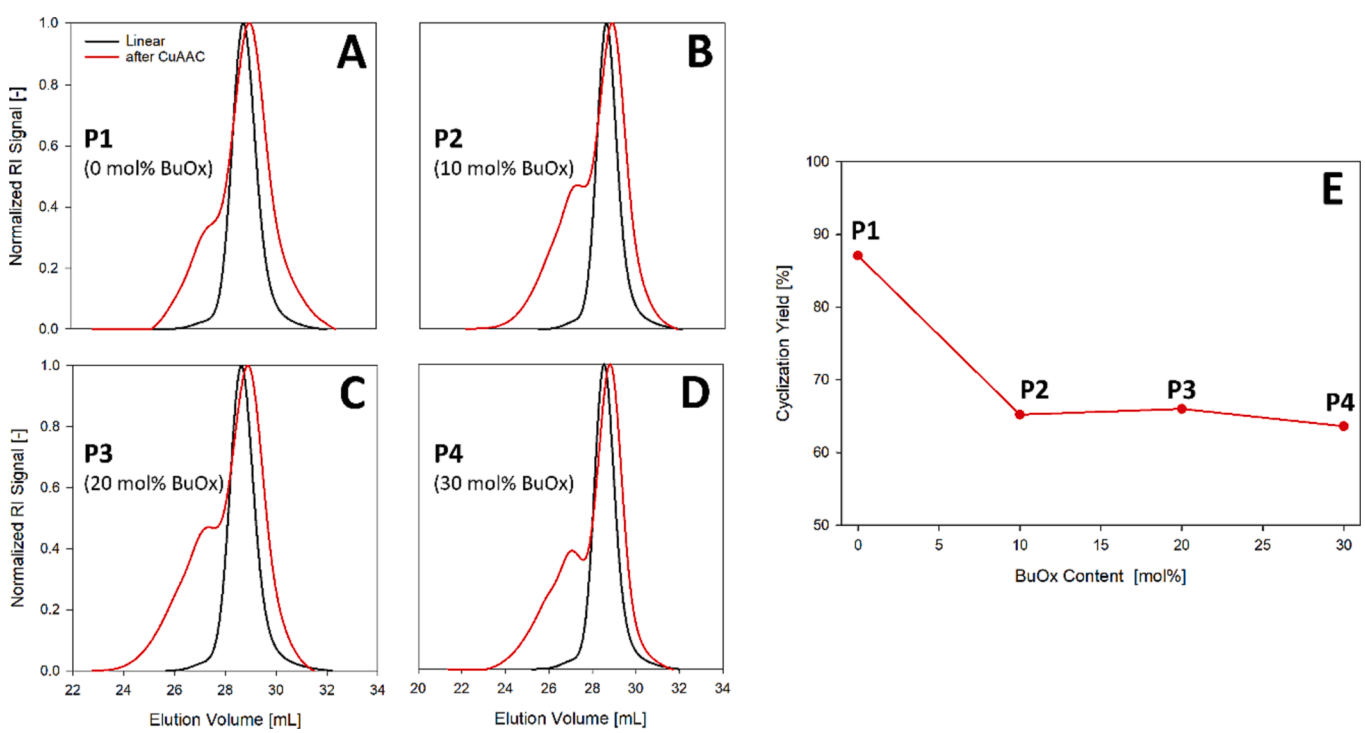


Fig. 2. SEC elugrams of **P1** (A), **P2** (B), **P3** (C) and **P4** (D) before (black) and after (red) CuAAC in deionized water at a polymer concentration of 5 mg mL^{-1}. To compare the results in a semi-quantitative way, the yield of the monocyclic product after CuAAC was determined (E). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

was approximated by relating the signal area of the monocyclic species in each elugram curve to the overall area of each elugram (Fig. S7, Equation S1). It should be noted that the values obtained by this semi-quantitative method should not be taken as absolute but rather serve as means to compare samples in this study relative to one another, as this approach is subject to several assumptions [27]. Firstly, the linear precursor polymer has been fully or mostly converted during CuAAC into either inter- or intrachain coupled products, leaving little or none of the SEC trace of the linear polymer in the final SEC trace post-CuAAC. Additionally, it is assumed that the cyclic species is displayed as a Gaussian curve in the elugram and that the RI detector response during SEC measurements increases similarly for each polymer species with increasing concentration.

The most hydrophilic polymer, **P1**, forms almost exclusively the monocyclic species with a total yield of 87 %. The high conversion of this highly solvated polymer into the cyclic species at the given concentration is consistent with the results obtained in our previous studies [21,44]. However, introducing 10 mol% BuOx into the polymer (**P2**), resulted in the cyclization yield decreasing significantly to 65 %, indicating that a small degree of hydrophobicity decreases polymer solvation sufficiently to affect the cyclization efficiency. This may lead to an increased collapse of the polymer chain in solution, ultimately favouring hydrophobic interactions between polymer chains and hence coupling reactions between the chain ends of several polymer chains, rather than intrachain cyclization. Additionally, as hypothesized in our previous study [44], the collapse of the polymer chain in solution could also lead to a decreased accessibility of the polymer end-groups for the Cu(I) catalyst in solution, hence hindering intramolecular ring closure and favouring more interchain coupling reactions between polymer chains.

Interestingly, however, the yields of the monocyclic product for **P3** and **P4**, containing 20 and 30 mol% BuOx, respectively, were similar to that achieved for **P2**. This is somewhat surprising, as a higher BuOx content in the polymer backbone is expected to lead to more interactions between the BuOx units within the polymer chain, and hence a change of the polymer conformation in solution into a more collapsed state. If catalyst accessibility was a main factor for the decrease in cyclization efficiency, the yields of the monocyclic product should change as the BuOx content increases and the polymer conformation in solution changes. As this is not the case, it seems more likely that the main cause for the decrease in cyclization efficiency from **P1** to **P2** originates from the previously mentioned aggregation of several polymer chains in solution through hydrophobic interactions, which ultimately leads to higher local concentrations of these polymers, and hence the formation of more interchain coupled products.

Furthermore, it should be noted that the decreased solvation of **P2-P4** in our study did not lead to an increase in cyclization efficiency compared to the hydrophilic analogue **P1**, in contrast to previous studies that investigated CuAAC cyclization of hydrophobic polymers in organic solvents [27,46]. A possible explanation for this discrepancy can be found in the polymer structure itself. While the mentioned studies used hydrophobic homopolymers that can fully collapse in poor solvents, we used amphiphilic copolymers of hydrophilic EtOx and hydrophobic BuOx, in which the hydrophilic units prevent the polymer chains from collapsing fully under the given reaction conditions, ultimately changing the reaction outcome. Additionally, we have not been able to emulate the quantitative cyclization yields as described by Zhang *et al.* [45] for any of the amphiphilic copolymers **P2-P4** under our given reaction conditions. A plausible reason for this is differences in the cyclization techniques. Zhang *et al.* used hydrophobic anthracene end-groups to form the monocyclic product, which favours encapsulation of these units into the hydrophobic core formed by the hydrophobic monomer units in the polymer chain. At the same time this minimizes the contact between end-groups of multiple polymer chains. This encapsulation is less likely to happen with the charged azide group used in CuAAC reactions. Additionally, the molecular weight of the polymers in the current study (<10 kg mol^{-1}) were significantly smaller than the ones in the mentioned study (>200 kg mol^{-1}), which would affect the size of the hydrophobic pocket, and hence change the efficiency of end-group encapsulation and ultimately the cyclization efficiency. Finally, at a constant mass-weighted polymer concentration in solution, as used in the current study and by Zhang *et al.*, the lower molecular weight polymers result in a higher number of individual polymer chains in solution compared to the higher-molecular weight analogues. This enhances the overall possibility of hydrophobic interactions between the low molecular weight polymers in the current study, favouring intermolecular reactions.

To monitor the conversion of the heterotelechelic precursor polymers (Fig. 3, black) into the respective CuAAC products (Fig. 3, red), SEC, as well as ^{1}H NMR and IR spectroscopy, were employed. As mentioned previously, a high conversion of all precursor polymers **P1-P4** is indicated by the significant signal shifts in the SEC elugrams. Beyond that, the ^{1}H NMR spectra for **P1-P4** showed a high conversion of the alkyne protons with a signal group between 4.05 and 4.35 ppm into the respective triazole species with a signal at approximately 3.60 and 4.60 ppm. This is consistent with previous reports on CuAAC reaction of POx [9,22,27]. However, it should be noted that a small amount of residual alkyne signal remained for all polymers post-CuAAC, suggesting the presence of some polymer chains without azide functionality. This suggestion was further supported with IR spectroscopy, as the azide signal of all polymers at about 2100 cm^{-1} was fully consumed, demonstrating the presence of more alkyne than azide moieties within the polymer chains. Generally, the presence of unreacted linear polymer precursors is undesired as it can alter the material properties [22,57] and interfere with cyclization yield determination. However, this was deemed acceptable in the current fundamental study as no material properties are evaluated. Furthermore, the residual alkyne signal in the ^{1}H NMR spectra after CuAAC was similar in all polymer batches, ensuring comparability of results between these systems. Semi-quantitative end-group analysis via integration of the alkyne signals in the ^{1}H NMR spectra was attempted to determine the exact percentage of polymers without azide group, however, the low signal-to-noise ratio in these areas leads only to inconclusive values. It should also be noted that these low amounts of unfunctionalized, and hence unreactive, linear precursor polymers led to some degree of signal broadening of the monocyclic species in the SEC elugrams.

Overall, our results highlight that the cyclization efficiency of POx (co)polymers is significantly affected by the net polymer hydrophobicity, which should be considered when synthesizing cyclic polymers. At a molecular weight of below 10 kg mol^{-1}, the yield of the monocyclic species dropped significantly upon introduction of as little as 10 mol% of hydrophobic BuOx units into the polymer chain. Further increase of the BuOx feed ratio only yielded marginally different results, indicating strong hydrophobic interactions between polymers in solution, even at 10 mol% BuOx content. All systems showed overall high conversion into their respective CuAAC species, hence confirming high polymer chain end fidelity. This was demonstrated through end-group analysis via ^{1}H NMR and IR spectroscopy, as well as SEC.

4.3. The influence of polymer concentration on cyclization efficiency

To evaluate how different polymer concentrations affect the cyclization efficiency, the most hydrophilic polymer **P1** (Fig. 4A) and the most hydrophobic polymer **P4** (Fig. 4B) were subjected to CuAAC conditions at different polymer concentrations in deionized water and evaluated for their cyclization efficiency.

Keeping the concentrations of the NaAsc and $CuSO_4$ catalysts the same (20 and 10 mM, respectively) but changing the polymer concentrations from 5 to 10 and finally 20 mg mL^{-1} led to an increase in interchain coupling product formation with increasing polymer concentration for both **P1** and **P4** (Fig. 4C). The largest decrease in cyclization efficiency was observed when increasing the polymer concentration from 5 to 10 mg mL^{-1}. Between these two experiments,

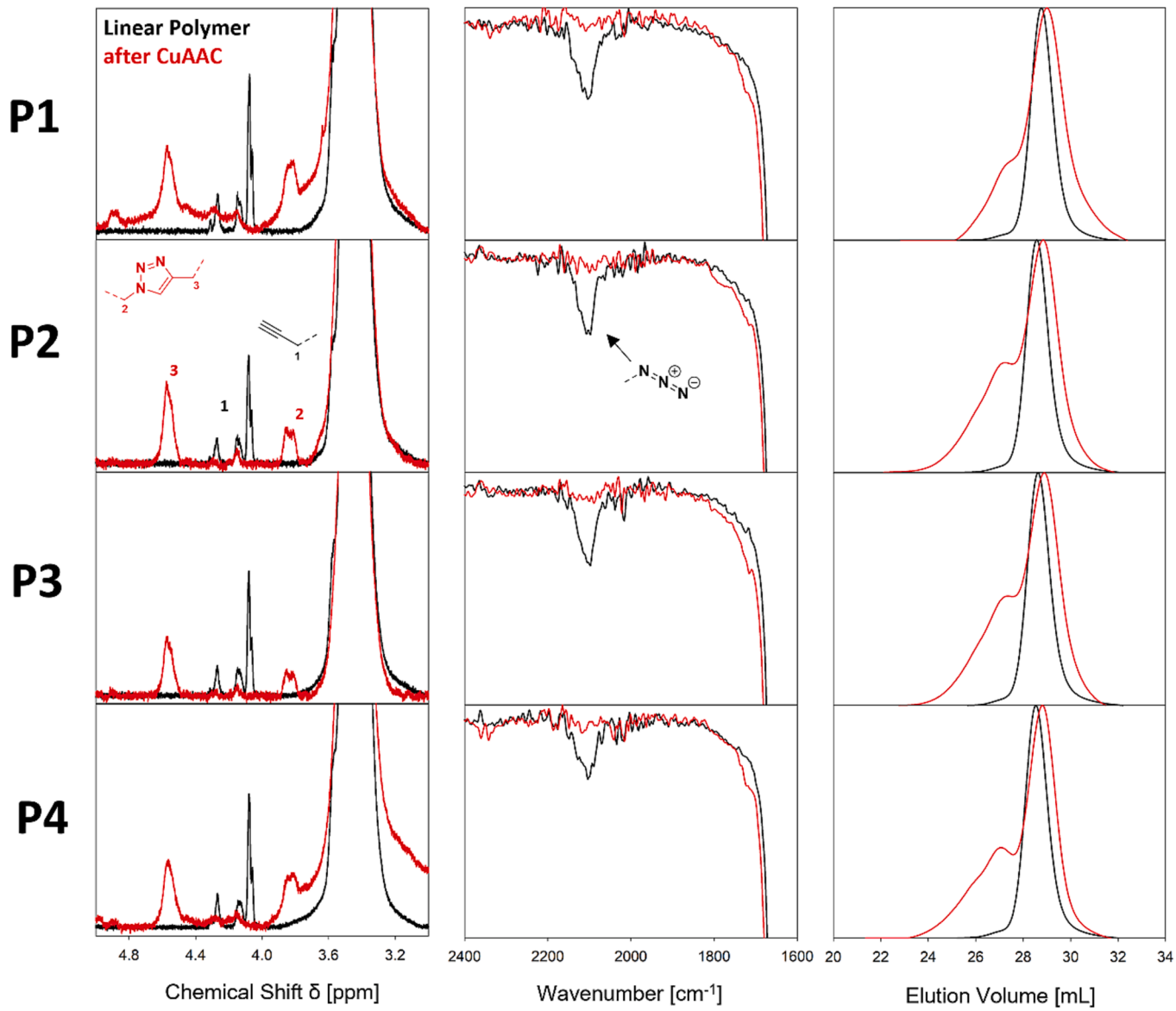


Fig. 3. ^{1}H NMR (left) and IR (middle) spectra as well as SEC elugrams (right) of **P1**-**P4** before (black) and after (red) CuAAC in deionized water at a polymer concentration of 5 mg mL^{-1}. NMR and IR spectra were cropped for better visibility of the relevant signals. Chemical structures included for clarification. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the yield of the monocyclic product decreased from 86 to 39 % for the hydrophilic **P1** and 63 to 41 % for the amphiphilic **P4**, indicating a strong dependence of the polymer concentration on the cyclization efficiency. Interestingly, the concentration had a greater influence of cyclization efficiency of **P1** compared to the BuOx-containing **P4**. Due to hydrophobic interactions between **P4** polymer chains in solution, it can be assumed that these interactions generally led to a higher local concentration of these polymers in solution even at lower polymer concentrations, increasing the chance for interchain coupling reactions. This effect is presumably amplified by increasing the number of polymer chains in solution, i.e., an increase in polymer concentration, to yield more interchain coupled products. However, due to the initial interactions between polymer chains, the difference in cyclization efficiency was less pronounced. The change in **P1**, on the other hand, was significantly larger, as the polymer chains are initially well-distributed and solvated in solution, favouring intramolecular ring closure. However, as the polymer concentration increased, interactions between polymers and closer proximity of polymer chains to each other significantly enhanced the chance of interchain coupling, resulting in a significant decrease in cyclization efficiency. Furthermore, it should be noted that at 10 mg mL^{-1}, and even 20 mg mL^{-1}, the cyclization efficiencies of **P1** and **P4** were similar, just below 40 %. Overall, these results highlight that the effect of polymer solvation on the cyclization efficiency diminishes as the polymer concentration increases due to strong interactions between polymer chains in solution independent of their amphiphilicity. However, especially at the lowest concentration of 5 mg mL^{-1} the difference between **P1** and **P4** was significant, indicating that polymer amphiphilicity and hence solvation could affect the cyclization process especially at lower polymer concentrations, as is the case for many current cyclization protocols that require high dilution [4,8,26,58,59].

4.4. *The influence of reaction temperature on cyclization efficiency*

Since the introduction of BuOx resulted in observable T_{CP}, it was of interest to evaluate the influence of reaction temperature during CuAAC (Fig. 5) on the cyclization yield. In this case, the hydrophilic **P1** (Fig. 5A) was compared to **P3** (Fig. 5B), containing 20 mol% BuOx. The T_{CP} of **P3**

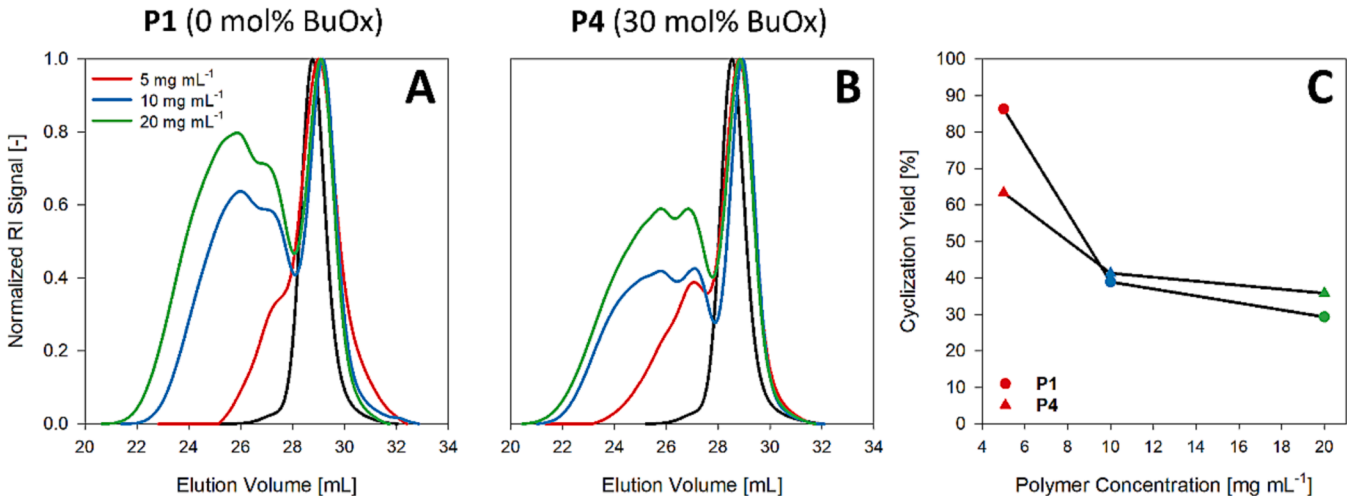


Fig. 4. SEC elugrams showing the CuAAC reaction products of **P1** (A) and **P4** (B) before (black) and after CuAAC at polymer concentrations of 5 (red), 10 (blue) and 20 mg mL^{-1} (green). Yields of the monocyclic product are shown for better comparison between samples (C). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

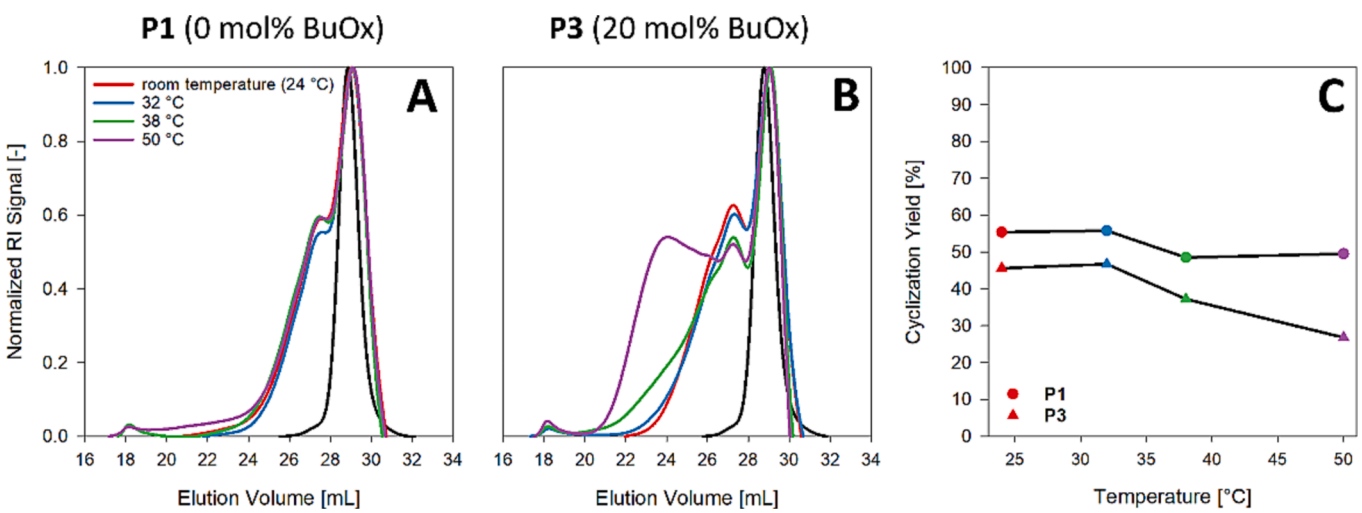


Fig. 5. SEC elugrams showing the CuAAC reaction products of **P1** (A) and **P3** (B) before (black) and after CuAAC at reaction temperatures of 24 (red), 32 (blue), 38 (green) and 50 °C (purple). In all experiments the polymer concentration was kept constant at 10 mg mL^{-1}. Yields of the monocyclic product are shown for better comparison between samples (C). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

was investigated using DLS at polymer concentrations of 5, 10 and 20 mg mL^{-1} in deionized water (Fig. S8) to determine a temperature that would allow sufficient temperature adjustments between the T_{CP}, at which the polymer starts precipitating out of solution, and room temperature. Based on the DLS results, a polymer concentration of 10 mg mL^{-1}, at which **P3** showed a T_{CP} of 42 °C, was chosen for the temperature study.

Both polymers were subjected to a total of four different reaction conditions. The first two experiments were conducted at 24 and 32 °C, at which **P1** and **P3** were both fully soluble in the CuAAC reaction mixture. These conditions aimed to isolate the influence of heating on the reaction outcome without inducing total collapse and precipitation of the thermoresponsive **P3**. For the third experiment, the reaction mixtures were heated to 38 °C. At this temperature, **P3** was still soluble in the reaction mixture, however, precipitation was induced through addition of the $CuSO_4$ catalyst, i.e., through salting-out as the CuAAC reaction started. For the fourth experiment at 50 °C **P3** was allowed to precipitate out of solution before addition of $CuSO_4$, thereby probing the influence of pre-formed polymer aggregates on the cyclization efficiency.

Surprisingly, the hydrophilic, well-solubilized **P1** formed a similar product mixture independent of the employed reaction temperature (Fig. 5C). Apart from the formation of small amounts of higher-order coupled products at 38 and 50 °C, neither the nature of the interchain coupled products nor the overall cyclization yield changed significantly, indicating only minor effects of the reaction temperature on the product formation. This result highlights that the common notion of heating CuAAC reactions [15,16,60] to maximize the cyclization yield may not be necessary to achieve the desired reaction outcome for well-solvated systems. Furthermore, these results are in line with the data obtained by a previous study on fatty-acid-substituted POx, cyclized via CuAAC in tetrahydrofuran/hexane mixtures [27]. While in their case the cyclization efficiency increased by 13 % when the reaction temperature was increased from 25 to 50 °C, further heating to 70 °C did not alter the composition of the product mixture. A possible explanation for this behaviour, as mentioned by the authors of this study, is that an increase in reaction temperature increases the rate of cyclization and interchain coupling reactions alike, leading to similar reaction outcomes. This argument seems valid for the hydrophilic **P1** homopolymer of this study, as the solvation of this polymer in water is maintained up to a temperature of 80 °C (Fig. 1, red), indicating that the polymer conformation is

not affected by increased temperatures and no polymer aggregation occurs.

A similar behaviour can be observed for **P3** when the temperature is increased from 24 to 32 °C, i.e., when the polymer was allowed to remain in solution during CuAAC. The difference in cyclization efficiency between these two temperature stages was marginal (46 % (24 °C) vs 47 % (32 °C)), showing that heating without affecting the overall solubility of the polymer chain in solution leads to comparable cyclic-to-oligomer product ratios. It also highlights that this effect occurs independent of the polymer amphiphilicity. The situation changed, however, as the T_{CP} of **P3** is reached. At a reaction temperature of 38 °C, where the polymer precipitated as soon as the CuAAC reaction was initiated by addition of the Cu(II) salt, the overall cyclization yield decreased from 47 to 37 %. Furthermore, higher-order interchain coupled species began to form, potentially due to the high polymer concentrations present in the precipitated aggregates. At a temperature of 50 °C, this effect intensified, due to excessive formation of polymer aggregates prior to the addition of the Cu(II) salt. This leads to a drop of the cyclization efficiency to 27 %, with the number of higher-order interchain coupled products being more prevalent. To show that the precipitation of **P3** out of solution did not decrease the overall conversion of the linear precursor polymer into CuAAC products, the product mixtures were analysed using ^{1}H NMR and IR spectroscopy (Fig. S9). However, no significant differences in precursor consumption could be observed in either experiment, indicating that **P3** is being converted into its CuAAC products, even when precipitated out of solution. Additionally, consistent with our previous experiments, the cyclization yield for the amphiphilic **P3** remained lower than for the corresponding hydrophilic analogue **P1**, independent of the temperature, highlighting that the polymer amphiphilicity and hence solvation is a useful handle to access higher yields of the monocyclic product.

Overall, our results indicate that the precipitation of the precursor polymer from solution decreases the overall cyclization efficiency independent of when the precipitation occurs. As CuAAC protocols require the use of salts to initiate the reaction, salting-out effects and overall polymer solubility should be considered when working with amphiphilic polymers and more generally in aqueous solvents.

5. Conclusion

In conclusion, we have been able to show that polymer amphiphilicity is a significant factor that affects cyclization efficiency of POx in water. Monocyclization was consistently more favoured than intermolecular coupling reactions for the hydrophilic **P1** compared to the increasingly more amphiphilic **P2-P4** polymers. This difference was more visible at lower polymer concentration. It is hypothesized that the lower cyclization efficiency in the amphiphilic systems can be attributed to hydrophobic interactions between BuOx units in solution leading to higher local polymer concentrations and hence an increased tendency to form interchain coupled oligomers rather than the intramolecularly reacted monocyclic species. At higher concentrations, the effect of the hydrophobic interactions between polymers was less prominent, due to the close proximity of polymer chains in solution.

Finally, the reaction temperature could be increased without observing significant changes in the CuAAC product mixture for all systems as long as the solubility of the polymer chain in solution was maintained. Formation of larger polymer aggregates and precipitation of the polymer out of solution at elevated temperatures was found to favour the formation of higher-order interchain coupled products due to high local polymer concentrations within the aggregates. Overall, we could demonstrate that polymer amphiphilicity is one of the governing factors affecting cyclization efficiency and hence should be considered when optimizing cyclization reaction conditions.

CRediT authorship contribution statement

Nick Huettner: Conceptualization, Methodology, Validation, Investigation, Data Curation, Writing – Original Draft, Writing -Review & Editing, Visualization. **Hendrik Frisch:** Supervision, Writing – original draft, Writing – review & editing. **Tim R. Dargaville:** Conceptualization, Methodology, Validation, Writing – Original Draft, Writing -Review & Editing, Supervision, Funding Acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2023.112659.

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